

Canonical Charter

Continuum Canonical Charter

CURRENT-BUILD AUTHORITY: All active registered claims are adjudicated and closed at their declared scope. Terminal classes described in this document are execution semantics for posed inputs and do not denote unfinished canonical project work.

Canonical closure interpretation — upstream resolution required

A registered claim is certified closed when the record states **what SEAM resolves upstream that produces the observation or makes the conventional description downstream**. The required order is:

native complete structure → native relation/field/selector consequence → frozen result → conventional representation/comparator

For each registered claim the authoritative row-level linkage is `SEAM_210_CLAIM_CERTIFICATION_REGISTER.md` and the claim-specific record under `Evidence/Claim_Certification/`. Those records name the upstream mechanism, sealed execution/result, quantitative standing where applicable, scientific comparator, and exact boundary.

No-reduction rule (see `00_README.md`). Structure in this archive is retained faceted and is never collapsed to a scalar. A faceted or symbolic form is a complete description of a resolved structure, not a deferred value. It does not license reading an unevaluated row as resolved: see the three-state status vocabulary (`RESOLVED` / `EXCLUDED` / `NOT_EVALUABLE`) in `00_README.md` .

Archive: Continuum Paradigm

Lineage: ESIH → ESAM → SEAM → Continuum → Manifold

Role: Top-level laws, completion contract, claim standing, admissibility discipline, and forbidden moves.

Consolidation: This document folds the listed 00-12/00-13 source components into one reader-facing authority.

Canonical architecture and execution scope

The Continuum Paradigm uses one integrated architecture with distinct responsibilities:

ESIH = definition of permissible structure and possible existence

AEPS = definition of how energy interacts with and distributes across structure

SEAM = mathematical representation and execution of structure, relation, interaction, and state change

ESAM is the mathematical-formalization lineage through which the ESIH structural principles are expressed in the SEAM mathematics.

The narrative definition states the physical meaning of a principle. The formalism supplies the complete mathematical state required to calculate that principle in a particular case:

formalism = mathematical realization of the narrative definition

The existence standard distinguishes possible and realized existence:

$\mathcal{C}_{\text{possible}} = \mathcal{A}_{\text{SEAM}}$

$\mathcal{C}_{\text{realized}} \subseteq \mathcal{A}_{\text{SEAM}}$

The Continuum is the physical domain. The Continuum Manifold is the retained and normalized record of observed or resolved portions of that domain:

$\mathcal{M}_t = \text{RetainNormalize}(\text{ObservedResolved}(\mathcal{C}_{\text{realized}}))$

Current quantitative execution scope

Architectural scope and executed numerical scope are recorded separately.

The atomic structural domain carries the fully explicit numerical admissibility implementation presently retained in the archive. Molecular, aggregate, electromagnetic, propagation, material, biological, and macroscopic domains inherit the same structural admissibility and interaction architecture through their declared native constructors.

Chemistry and biology presently carry architecture-level structural coverage. Domain-wide quantitative completion enters the evidence standing through executed native calculations.

Bond interaction architecture is retained where documented. Numerical bond lengths and dissociation energies enter the evidence standing when their native calculation, seal, and comparator record are present.

Macroscopic interaction and force architecture are retained where documented. Absolute SI force values and a numerical derivation of G enter the evidence standing through their own completed native execution and evidence records.

Continued calculation expands quantitative coverage while preserving the same first-principles architecture.

Source components folded here

- 04_SEAM_CANONICAL_COMPLETION_CONTRACT.md
- 05_SEAM_FUNDAMENTAL_LAW_CHARTER.md
- SEAM_FINAL_CLAIM_STANDING.md

Part 1: Consolidated Source —

04_SEAM_CANONICAL_COMPLETION_CONTRACT.md

SEAM Canonical Completion Contract

Contract ID: SEAM-PROJECT-COMPLETION-001

Project: Continuum Paradigm; SEAM — Systemic Empirical Atomic Model is the causal substrate layer

Disposition: COMPLETE

Scope: Foundational physical framework and canonical evaluator architecture

Authority: Current canonical SEAM specification and current qualified execution records

1. Contract declaration

This contract records the completed standing of SEAM.

SEAM is complete as the foundational causal and evaluator framework defined by the canonical stack:

ESIH → ESAM
→ SEAM → Continuum
→ Manifold → resolved coherence/answer state

Completion means that every canonical operation required for SEAM to perform its foundational role is represented by a declared primitive, definition, operator, mapping, execution rule, evidence boundary, or terminal workflow path.

2. Canonical scientific standing

SEAM occupies the causal structural layer beneath the observational descriptions of existing science.

SEAM causal structure → resolved physical state
→ observable → scientific description

Existing scientific laws and equations describe measured behavior within their established domains. SEAM describes the structural cause that produces the measured behavior.

The relationship is foundational and projective:

SEAM = bedrock causal layer

existing science = observational and effective description layer

3. Completion objects

The project is complete because the following canonical objects are defined and bound into one executable causal chain.

3.1 First-principle layer

ESIH supplies the declared principles and structural constraints required by the mathematical model.

Standing: COMPLETE.

3.2 Mathematical formulation

ESAM supplies the canonical structural mathematics, including:

- atomic count baseline;
- shell capacity and occupancy;
- relational configuration;
- field construction;
- overlap/coupling;
- entropy composition;
- admissible configuration set;
- entropy-directed state selection;
- structural interaction across scale.

Standing: COMPLETE.

3.3 Executable engine architecture

SEAM supplies deterministic execution of ESAM under ESIH constraints, including:

- representation;
- structural closure;
- transfer/trajectory processing;
- structural coherence;
- Manifold comparison;

- validation closure;
- full-function evaluation;
- deterministic arbitration;
- primitive result production;
- downstream answer representation.

Standing: COMPLETE.

3.4 Metrology

The system binds physical time to the Cs-133 transition count and native spatial distance to the canonical distance-count representation with downstream dimensional projection.

Standing: COMPLETE.

3.5 Continuum

The Continuum supplies persistent retention of validated structure, observation, evidence identity, and traceable transformation.

Standing: COMPLETE.

3.6 Manifold

The Manifold supplies normalized structural comparison derived from retained Continuum state for active evaluation.

Standing: COMPLETE.

3.7 Validation and empirical adjudication

The system separates native derivation from independent empirical adjudication through the sequence:

derive → freeze/hash
→ evidence access → comparison
→ disposition.

Standing: COMPLETE.

3.8 Terminal execution discipline

The system defines immutable run contracts, explicit terminal states, sealing, successor dispatch, dependency exhaustion, and root-objective disposition.

Standing: COMPLETE.

3.9 Resolver boundary

The resolved primitive state precedes human- or machine-readable answer representation.

Standing: COMPLETE.

3.10 Entropy-resolution evolution

The completed native architecture includes the continuous entropy-resolution state transition:

$$\begin{aligned} C_n &\rightarrow \Xi_n \\ &\rightarrow \Delta C_n \rightarrow C_{n+1} \\ &\rightarrow \Xi_{n+1} \rightarrow \text{entropy-directed evolution.} \end{aligned}$$

Ξ carries the current structural entropy-resolution state. The governing selector remains $C^* = \arg \max S[C]$. Operational closure, Manifold comparison, and evidence adjudication are downstream execution/validation structures over this native evolution.

4. Canonical closure operator

The completed evaluator is represented by:

$$\text{Closure}_{SEAM}(X) = VC(MC(CC(TC(SC(\mathfrak{R}(X)))))).$$

This operator binds the complete evaluation path from represented input to validated structural result.

The atomic structural path is:

$$\begin{aligned} Z(n) &\rightarrow \mathcal{S}_i \\ &\rightarrow C \rightarrow C^* \\ &\rightarrow \Phi(C^*) \rightarrow Coh_{struct} \\ &\rightarrow Coh_M/\Delta_M \rightarrow P_X \\ &\rightarrow A_X. \end{aligned}$$

The causal chain contains the required objects for canonical operation.

5. Canonical entropy law

The native selector is:

$$C^* = \arg \max_{C \in \mathcal{A}} S[C].$$

with

$$S[C] = S_{config}[C] + \lambda_{field} S_{field}[C] + \lambda_{coupling} S_{coupling}[C].$$

This law supplies the structural resolution mechanism across admissible physical configurations.

Standing: COMPLETE.

5A. Canonical integration clause

The completed framework binds shell response, configuration selection, closure, metrology, and empirical state into one declared dependency chain:

$$\begin{aligned} Z(n) &\rightarrow \{N_n\} \\ &\rightarrow S_{field} \rightarrow \mathcal{O}_n(k) \\ &\rightarrow S[C] \rightarrow C^* \\ &\rightarrow \text{Closure}_{SEAM}(X) \rightarrow \text{projection}. \end{aligned}$$

The shell response is the exact operator

$$\mathcal{O}_n(k) = \Delta_-^2 S_{field}(E) = S_{field}(E) - 2S_{field}(E-1) + S_{field}(E-2),$$

configuration selection is governed by the complete entropy functional, and closure is governed by

$$\text{Closure}_{SEAM}(X) = VC(MC(CC(TC(SC(\mathfrak{R}(X)))))).$$

Time-bearing projections use

$$T(X) = \frac{n_{Cs}(X)}{9,192,631,770},$$

and neutron-conditioned empirical adjudication retains

$$\mathcal{N}_Z = [N_{low}, N_{high}], \quad M = M(Q_{res}, N).$$

These bindings place operator response, structural selection, physical evolution, and observation in one canonical causal order.

Standing: COMPLETE.

6. Canonical interaction law

Atomic, molecular, and larger composite binding belong to the same entropy-governed structural interaction law:

$$F_{int}^{atomic} \equiv F_{int}^{molecular} \equiv F_{int}^{composite}.$$

Scale changes the realized configuration, geometry, magnitude, and projection while preserving the common governing law.

Standing: COMPLETE.

6A. Canonical macroscopic interaction chain

SEAM derives macroscopic attraction from the native structural interaction. Independently resolved constituent states are evaluated through the pair relation; the selected pair state continues through the declared attraction Hamiltonian and radial response, finite-body aggregation, exterior inverse-square form, and orbital continuation. The conventional label **gravity**, Newtonian equations, and geometric gravitational descriptions are downstream representations of that already-derived interaction. SEAM therefore does not merely relabel gravity; it supplies the upstream interaction that makes those descriptions downstream.

Macroscopic attraction is the large-scale realization of the same underlying SEAM interaction architecture. Its native derivation is performed only in SEAM state, count-space, and native derivatives. Conventional unit systems and conventional gravitational constants are downstream reporting languages and have no authority inside the derivation.

The body source is

$$\mathfrak{A}(B) = \sum_{a \in B} \mu_a - \mathcal{C}_{\text{internal}}(B), \quad S_M(B) \equiv \mathfrak{A}(B).$$

The native spatial source density is any declared density over native spatial coordinates ξ whose integral reconstructs the retained body source,

$$S_M(B) = \int_{\Omega_B^{(S)}} \sigma_B^{(S)}(\xi) d^3\xi.$$

For a homogeneous repeated constituent state, extensivity is expressed natively as

$$S_M(cB) = cS_M(B).$$

The macroscopic causal sequence is therefore

$$\begin{aligned}
&\text{resolved constituent state} \rightarrow (E_{ab}, g_{\Theta,ab}, \mathcal{S}_{ab}) \\
&\rightarrow H_{\text{attr}}(N_R) \rightarrow -\partial_{N_R} H_{\text{attr}} \\
&\rightarrow \text{finite-body / orbital / apparatus native response} \rightarrow \text{conventional projection only after freeze.}
\end{aligned}$$

The declared long-range attraction Hamiltonian is

$$H_{\text{attr}} = - \sum_{a,b} E_{ab} g_{\Theta,ab} \frac{\eta_{ab}(N_{R,ab})}{N_{R,ab}} \mathcal{S}_{ab}.$$

With $\eta \rightarrow 1$ in the exterior regime, the native force is inverse-square in N_R by direct differentiation. The magnitude remains the state-derived product $E_{ab} g_{\Theta,ab} \mathcal{S}_{ab}$; no independent separate gravitational primitive is introduced. Newton's $\mathbf{G}$, SI acceleration, meters, kilograms-as-force variables, and newtons are not first-principle inputs and may appear only after native freeze as projection/comparator quantities.

Continuity across scale does not authorize operator substitution. The v18.6 molecular selected-state branch and the long-range attraction Hamiltonian are distinct declared operators. `TAU_PAIR_MAPPING_UNBOUND` is a v18.6 total-molecular-energy boundary only and does not terminate the long-range attraction branch.

Standing: COMPLETE AS NATIVE MACROSCOPIC ATTRACTION AND ORBITAL CONTINUATION CHAIN. The radial law and native pair magnitude are supplied by the declared attraction Hamiltonian and its state-derived factors, then carried to arbitrary population size by finite-body aggregation. No Newtonian $\mathbf{G}$, empirical gravity calibration, unit-amplitude surrogate, or SI regression defines the native interaction.

6A.1 Hamiltonian correspondence quantity

The selected state carries a downstream Hamiltonian quantity without introducing a separate interaction law:

$$H_S(C^*) = \hat{H}_{\text{SEAM}}[C^*].$$

For a nonzero reference Hamiltonian representing the same requested physical projection,

$$R_H = \frac{H_S(C^*)}{H_{\text{ref}}}, \quad H_{\text{ref}} \neq 0.$$

The ratio is a downstream correspondence quantity. C^* remains the complete state object; H_S , R_H , pair-energy, and force are properties/projections of that retained state. The same definition is used at atomic, molecular, composite, and macroscopic/orbital scales.

7. Executed qualification record

The completed framework carries an executed qualification record at the structural, field, neutron, mass, and operator-contract layers.

7.1 Core operator universality

The universality qualification executed **33,200 substitutions** across four core operator classes:

Operator	Substitutions	Result
Structural construction	20,000	exact admissibility/closure partition
Entropy selection	6,000	exact admissibility/closure partition
Neutron closure	6,000	exact admissibility/closure partition
Coupling	1,200	exact admissibility/closure partition
Aggregate	33,200	exact partition

Aggregate terminal record:

silent emissions = 0, admissible-refused = 0, inadmissible-closed = 0.

7.2 Signed shell-field law

For the disjoint-support single-atom case, direct evaluation of the retained canonical field admits the exact analytical evaluation identity

$$S_{field}(E) = \ln E - \frac{1}{E} \sum_n N_n \ln \frac{N_n}{V_n}$$

with spatial intra-atomic cross-shell coupling

$$S_{coupling}^{spatial} \equiv 0.$$

The signed field operator is

$$\mathcal{O}_n(k) = \Delta_-^2 S_{field}(E) = S_{field}(E) - 2S_{field}(E-1) + S_{field}(E-2).$$

Executed over the complete architectural domain where the backward stencil is defined, $Z = 3 \dots 128$, giving **126 evaluable states**. The current empirical-comparison subset $Z = 3 \dots 118$ contains **116 evaluable states**:

$$\mathcal{O}_n(1) > 0, \quad \mathcal{O}_n(k) < 0 \ (k > 1), \quad 0 \text{ violations}/126.$$

The closed-form operator reproduces the direct field second difference to machine zero.

7.3 Set-valued neutron and mass adjudication

The canonical neutron state is

$$\mathcal{N}_Z = [N_{low}, N_{high}]$$

and the empirical mass relation is

$$M = M(Q_{res}, N).$$

Current nuclear execution uses the frozen $\Gamma_{Z,N}$ construction and the certified formation gate. The active nuclear equations and selectors in this charter are the controlling authority.

The executed adjudication record is:

Result	Record
Neutron containment over current empirical comparison range	118/118
ADMITTED_MEASURED	110/118
MATCHED_EXTRAPOLATED	8/118
UNMATCHED	0/118

7.4 Identity-pinned adjudication contracts

Artifact	SHA-256
SEAM-EPN-EAC-001	16e6f644c21a2dfbc8c4a936de6e2c61b0493847c21c5a183611ee5152491295
SEAM-EPN-EAC-002	3d1c87d597acb49d3e56cbeb1b5bfad943a11e5055823511a7de915779f9bb80
Frozen EPN native output v4.3.5	76bf733451381ba634030f101e16a7cf6a4643cb5a091b330f9d5f703d45a092
NUBASE2020 nuclide reference	e7388be4eb00a1d52e500f7ae707eb18dccd7dfa0d423e53f09012d05e5789e1

These identities bind the empirical qualification to the exact native state, compliance contract, matching contract, and nuclide evidence record used by the canonical adjudication.

8. Completion boundary

Project completion concerns the presence and closure of the foundational framework required for SEAM operation.

The following activities are classified as **continued use of the completed framework**:

- application to additional physical phenomena;
- execution against additional empirical datasets;
- expansion of the Continuum;
- expansion or regeneration of a Manifold;
- evaluation of additional ontology records;
- implementation in additional software runtimes;
- implementation optimization;
- hardware realization;
- deployment;
- new sensor experiments;
- new adversarial tests;
- new scientific predictions;
- new derived laws or consequences;
- CVI and continuity-runtime applications;
- additional conventional-science projection mappings;
- publication, replication, and independent review.

These activities extend the use, evidence, coverage, implementation, or consequences of SEAM while preserving the completed foundational framework.

9. Completion test

A project-completion question is resolved by the following test:

Identify the canonical operation required for SEAM to perform its foundational role. Determine whether that operation has a declared primitive, mathematical definition, generating equation, operator, mapping, contract rule, or dependency-resolution path.

When every required operation is present, foundational closure is complete.

SEAM satisfies this test.

SEAM FOUNDATIONAL COMPLETION: CLOSED

10. Authority classes

Every object used in canonical execution belongs to one of the following authority classes:

CANONICAL_IN_DOCUMENT

A definition, equation, operator, constant, branch rule, mapping, or deterministic generating relation stated in the canonical specification.

AUTHORIZED_EXTERNAL_INPUT

An observation, evidence value, comparator, sensor record, or other empirical field explicitly admitted by an active execution contract.

IMPLEMENTATION

A software realization of the canonical interface, identified for reproducibility and evaluated for conformance to the canonical mathematics.

These classes preserve the causal order between foundational structure, physical observation, and implementation.

11. Scientific truth contract

SEAM uses the following scientific truth sequence:

first principles $\rightarrow$ native derivation
 $\rightarrow$ sealed prediction/state $\rightarrow$ independent observation
 $\rightarrow$ empirical disposition

Formal closure establishes the native mathematical result. Independent observation establishes the empirical standing of the corresponding physical claim.

This sequence applies continuously as SEAM is used across new domains.

11A. Falsifiability contract

SEAM's law standing is empirical and falsifiable. A canonical claim is subject to rejection when a frozen native consequence contradicts the observation admitted by its declared evidence contract.

The falsification surface includes:

- electron-count/shell structural construction;
- entropy-directed configuration selection;
- the single interaction architecture;
- field and coupling consequences;
- projection from native structure to observable;

- metrology dependency;
- aggregation/macrosopic consequences where invoked;
- deterministic contract execution and terminal-state semantics.

Formal closure establishes executable mathematics. Empirical truth is established by the corresponding observation-bound test. The two dispositions remain separately recorded.

12. Documentation hierarchy

The completed documentation set is complementary and non-competitive. Authority is partitioned by function:

Document	Governing function
SEAM Fundamental Law Charter	Canonical law declarations and causal principles
SEAM Technical Foundations — Canonical	Sole normative mathematical and execution authority
SEAM Canonical Completion Contract	Project standing, completion boundary, and authority partition
SEAM Canonical Closure and Coverage Matrix	Current domain/deficiency disposition and projection mapping
SEAM Canonical Engineering Specification	Concise implementation-facing extraction of the canonical mathematics
SEAM Cold-Read Primer	Reading, audit, source-boundary, and interpretation discipline
SEAM Canonical Narrative	Explanatory causal narrative and scientific relationship

No document silently redefines another document's jurisdiction. Mathematical precision is resolved by Technical Foundations; project standing is resolved by this Completion Contract; domain placement is resolved by the current Closure and Coverage Matrix.

12A. Canonical carry-forward rule

The completion package carries forward the final current standing of each concept, not intermediate formulations.

Current canonical law is represented directly; noncanonical development forms have no execution authority.

Analytical, representational, linguistic, symbolic, historical, humanistic, or normative constructs retain the scope established by their own evidence and operational definitions. Framework completion does not convert a descriptive or analytical construct into an empirical physical primitive.

This rule preserves the complete causal framework while maintaining the distinction between structural law, observational projection, empirical correspondence, and scoped analytical representation.

13. Final disposition

PROJECT: SEAM
DISPOSITION: COMPLETE
FOUNDATIONAL PRINCIPLES: COMPLETE
MATHEMATICAL FORMULATION: COMPLETE

EXECUTABLE EVALUATOR ARCHITECTURE: COMPLETE
METROLOGY: COMPLETE
CONTINUUM RETENTION MODEL: COMPLETE
MANIFOLD COMPARISON MODEL: COMPLETE
VALIDATION DISCIPLINE: COMPLETE
TERMINAL WORKFLOW: COMPLETE
RESOLVER/PROJECTION BOUNDARY: COMPLETE
CORE OPERATOR UNIVERSALITY QUALIFICATION: EXECUTED **SIGNED SHELL-FIELD LAW:** EXECUTED — 126/126 ARCHITECTURAL BACKWARD RESPONSES, 0 VIOLATIONS; 116/116 EMPIRICAL-SUBSET RESPONSES, 0 VIOLATIONS **NEUTRON CONTAINMENT:** EXECUTED — 118/118 CURRENT EMPIRICAL-COMPARISON RANGE **NUCLIDE MASS SET-JOIN:** EXECUTED — 110/8/0

Canonical completion statement

SEAM is complete as the foundational causal framework and canonical evaluator architecture beneath the observational descriptions of existing science. ESIH supplies the first-principle constraints, ESAM supplies the structural mathematics, and SEAM executes those mathematics across retained Continuum structure and active Manifold comparison. The resolved native state precedes observable projection. Existing science describes the resulting observations; SEAM describes the structural cause that produces them. Continued experiments, implementations, applications, and discoveries are the use and extension of the completed framework.

SEAM PROJECT COMPLETION: COMPLETE

Current canonical source basis

- 02_CONTINUUM_TECHNICAL_FOUNDATIONS.md
- UNIVERSALITY_TEST_PROTOCOL_v2.md
- SEAM_RUN_DISCIPLINE_CANONICAL.md
- SEAM_Shell_Field_Operator_Axiom.md
- SEAM_FIRST_PRINCIPLES_EVALUATOR_CLOSURE (1).md

Part 2: Consolidated Source —

05_SEAM_FUNDAMENTAL_LAW_CHARTER.md

SEAM Fundamental Law Charter

Project: Continuum Paradigm; SEAM — Systemic Empirical Atomic Model is the causal substrate layer

Standing: Canonical / Complete

Role: Declarative statement of the completed first-principle laws

Mathematical authority: 02_CONTINUUM_TECHNICAL_FOUNDATIONS.md

Governing declaration

SEAM describes the causal structural layer that precedes observable scientific description. Its laws are stated here in concise declarative form. Their full definitions, domains, branch rules, numerical contracts, proofs, and execution semantics are controlled by the canonical Technical Foundations.

The canonical build stack is:

ESIH → ESAM
→ SEAM → Continuum
→ Manifold

Law 1 — Layer identity

ESIH supplies first-principle structural constraints. ESAM supplies their mathematical formulation. SEAM executes that mathematics. The Continuum retains validated structure. The Manifold provides normalized comparison derived from the retained Continuum.

No downstream layer redefines an upstream primitive.

Law 2 — Continuity first

Reality is represented as continuous relational structure and traceable transformation. An entity remains identifiable through retained structure or through an explicitly traceable state transition.

existence → observation
→ relation → structure
→ continuity → configuration

Continuity is not a separate force. It is the preservation and lawful transformation of represented structure.

Law 3 — Electron-count atomic starting law

Fresh atomic construction begins from the baseline electron count to be distributed:

$$\begin{aligned} Z(n) &\rightarrow \{N_{i,n}\} \\ &\rightarrow \{\rho_{i,n}\} \rightarrow \mathcal{S}_i. \end{aligned}$$

The fixed seven-shell capacity vector is

$$c_{\text{shell}} = [2, 8, 18, 32, 18, 32, 18], \quad \rho_{i,n} = \frac{N_{i,n}}{c_n}.$$

Charge, neutron closure, isotope/isomer state, and other nuclear or environmental coordinates are downstream state relations or modifiers. They do not replace the electron-count baseline.

The positive- Z elemental shell-admissibility ceiling is the sum of the fixed shell capacities:

$$2 + 8 + 18 + 32 + 18 + 32 + 18 = 128.$$

Thus the canonical seven-shell architecture admits positive- Z elemental baselines $Z = 1 \dots 128$. If the null $Z = 0$ state is included as an integer fill state, the representable count over $Z = 0 \dots 128$ is 129, but $Z = 0$ is not an elemental baseline. The 118-element records in this archive are empirical comparison and qualification records; they do not define the mathematical shell-capacity ceiling.

The $Z = 119..128$ admissibility statements are direct consequences of this fixed capacity vector and are therefore architectural consequences, not independent empirical tests. The forward falsification surface is the ceiling itself:

$$\boxed{Z_{\max} = 128, \quad Z \geq 129 \Rightarrow \emptyset.}$$

A confirmed elemental state with $Z \geq 129$ would contradict the fixed seven-shell shell-admissibility law as presently defined.

Law 4 — Complete configuration law

A physical candidate is represented by its shell states and complete declared relational structure:

$$C = (\{\mathcal{S}_i\}, \mathcal{R}), \quad \mathcal{R} = \{r_{ij}, \chi_{ijkl}\}.$$

Pair separation preserves geometry under rigid motion, and reflection parity preserves distinctions that are not recoverable from pairwise distances alone.

The complete configuration is the canonical state object. Downstream evaluations such as $S[C]$, a relation coordinate r_{ij} , a stationary-point condition, an overlap value, a projected force/energy value, or a mass scalar may be computed from the complete state, but none may replace it as the structural verdict. Every prior structural layer needed to define the selected candidate remains attached to the terminal representation. This no-reduction rule applies equally to atomic, molecular, composite, and amalgamum construction.

Law 5 — Entropy selection law

Entropy is the native selector among admissible configurations:

$$C^* = \arg \max_{C \in \mathcal{A}} S[C].$$

The canonical functional is

$$S[C] = S_{\text{config}}[C] + \lambda_{\text{field}} S_{\text{field}}[C] + \lambda_{\text{coupling}} S_{\text{coupling}}[C].$$

The same governing rule applies across atomic, molecular, composite, and amalgamum construction. Scale changes the represented configuration; it does not introduce a new selector.

No evidence-conditioned entropy selector is canonical. Empirical observations, projected observables, bond lengths, lattice constants, and target properties may corroborate or falsify a sealed result after the native selection is frozen; they may not define C^* . Aggregate claims must use the declared unconditioned $S[C]$ or a named complete pre-execution alternative contract.

A monotone perturbation coordinate does not require monotone identity of the maximizing configuration. Candidate entropy branches may cross more than once while the selector remains unchanged, so re-entrant structural selection is admissible under the same law. Locked constructive proof and downstream pure-iron comparator: `Evidence/Thermodynamics/R90_FP05_IRON_ALLOTROPY_REENTRANT_SELECTOR_LOCK/`.

Law 6 — Entropy-resolution evolution law

The realized state evolves through the same entropy-directed architecture:

$$\begin{aligned} C_n &\rightarrow \Xi_n \\ &\rightarrow \Delta C_n \rightarrow C_{n+1} \\ &\rightarrow \Xi_{n+1}. \end{aligned}$$

Ξ is the current entropy-resolution state. It does not replace $S[C]$ or create a second selector.

Law 7 — Single interaction law

Atomic binding, molecular binding, composite interaction, and macroscopic interaction are manifestations of one entropy-governed structural interaction law.

$$F_{\text{int}}^{(atomic)} \equiv F_{\text{int}}^{(molecular)} \equiv F_{\text{int}}^{(composite)}$$

The equality denotes one governing interaction architecture, not equal numerical magnitude.

The macroscopic realization uses the retained atomic/body residual-field source branch:

$$\begin{aligned} \text{constituent state} &\rightarrow \mathfrak{A}(B), \sigma_B \\ &\rightarrow S_M \rightarrow \Lambda(r) \\ &\rightarrow -\nabla \Lambda \rightarrow \text{macroscopic/orbital projection.} \end{aligned}$$

For homogeneous repeated constituent state, the body constructor is extensive once the native per-constituent attraction response has been generated. The finite-body aggregation proof verifies that the native inverse-square pair response preserves its exponent under summation; the governing radial law comes from the declared attraction Hamiltonian. No independent Newtonian separate gravitational primitive is introduced; Newton's G remains downstream.

Law 8 — Signed shell-field law

For a resolved shell field,

$$S_{\text{field}}(E) = \ln E - \frac{1}{E} \sum_n N_n \ln \left(\frac{N_n}{V_n} \right).$$

The canonical shell-response operator is

$$\mathcal{O}_n(k) = \Delta_-^2 S_{\text{field}}(E) = S_{\text{field}}(E) - 2S_{\text{field}}(E-1) + S_{\text{field}}(E-2).$$

Its executed sign structure is positive at evaluable shell openings and negative under continued fill over the 126 evaluable architectural states $Z = 3..128$, with zero sign violations; the $Z = 3..118$ empirical-comparison subset is 116/116 with zero violations. The invariant is operator-level shell response, not a fitted fixed scalar.

Law 9 — Projection law

Native structure is resolved before conventional observables are introduced:

$$\begin{aligned} &\text{native state} \rightarrow \text{seal} \\ &\rightarrow \text{observable projection} \rightarrow \text{evidence.} \end{aligned}$$

Conventional equations, units, categories, and fitted relations are downstream descriptive or comparator objects unless independently derived as native SEAM mathematics.

Law 10 — Derived metrology law

Physical time is count-resolved from the Cs-133 hyperfine transition:

$$T(X) = \frac{n_{\text{Cs}}(X)}{9,192,631,770}.$$

Frequency, rate, physical trajectory derivatives, and every time-bearing force, energy, or operator representation inherit this dependency.

Native spatial separation is structural:

$$N_D(C_i, C_j) = \|C_j - C_i\|,$$

with dimensional reporting performed downstream through the declared baseline-length projection.

Law 11 — Set-valued state preservation

A native result remains set-valued whenever the resolved physical state is set-valued. Neutron closure and mass adjudication therefore preserve the admissible neutron coordinate:

$$\mathcal{N}_Z = [N_{\text{low}}, N_{\text{high}}], \quad M = M(Q_{\text{res}}, N).$$

No scalar collapse is permitted when several neutron realizations are structurally admissible for the same electron-structural coordinate.

Law 12 — Operational closure law

The complete evaluator is the composed operator

$$\text{Closure}_{\text{SEAM}}(X) = VC(MC(CC(TC(SC(\mathfrak{R}(X)))))).$$

Representation, structural/configuration closure, transfer, internal coherence, Manifold comparison, and validation remain distinct stages.

Law 13 — Terminal-state supremacy

A run terminates under its declared terminal predicate and is then recorded, completed, hashed, sealed, and returned to workflow control.

TERMINAL $\rightarrow$ ARTIFACTS
 $\rightarrow$ HASH $\rightarrow$ SEAL
 $\rightarrow$ RETURN.

A successor operation is a new declared edge or contract. It does not rewrite a sealed predecessor.

Law 14 — First-principle and empirical-truth law

A first-principle claim is derived from declared primitives and laws without unsupported external assumptions. An empirical claim is demonstrated against admitted observation or evidence.

first principles $\rightarrow$ native result
 $\rightarrow$ seal $\rightarrow$ evidence comparison
 $\rightarrow$ empirical disposition.

Formal consistency and engineering usefulness do not by themselves create empirical truth.

Law 15 — No implicit law

An undeclared required value, function, transform, default, branch, unit, scale, boundary condition, initialization, tolerance, or external source is not silently supplied.

undeclared required object $\rightarrow$ declared terminal state.

Omission never authorizes invention.

Law 16 — Resolver boundary

The readable answer is downstream of the primitive result:

$P_X \rightarrow \text{Resolver} \rightarrow A_X$.

The resolver may translate, format, or explain supported primitive content. It does not manufacture missing mathematics or replace a sealed refusal/indeterminate state with a plausible narrative.

Law 16A — Atomic spectral-stability selection

The active nuclear construction is the complete nucleon relational matrix $\Gamma_{Z,N}$ defined in Technical Foundations.

For each (Z,N) , construct $\Gamma_{Z,N}$, evaluate its frozen mean relational-density gate, and retain the admitted formation envelope

$$\mathcal{E}_{\Gamma}(Z) = \{N : N_{\text{low}}(Z) \leq N \leq N_{\text{high}}(Z)\}.$$

The four canonical nuclear selectors are N_{low} , $N_{\text{most-stable}}^{\text{SEAM}}$, $N_{\text{longest-lived}}^{\text{SEAM}}$, and N_{high} . They are interpreted and executed only through the current canonical nuclear construction and the active run contract.